delivery in the quarantined areas to contain the spread of the novel coronavirus. China was the first to adapt and co-opt drones to enforce the world's largest quarantine exercise. The drone's software can be modified to enforce preventive measures and increase infection detection and crowd management, reducing the risks of widespread surveillance and law enforcement.

Robots are another new technology being used to control the spread of Covid-19. These have been used to provide services and care for those quarantined or practising social distancing right from the time of the initial outbreak of the pandemic in China to its spread across the globe. Scientists and engineers have fast-tracked the 'testing' of robots and drones in public. Robots will help all the agencies who are seeking the most expedient and safest way to grapple with the outbreak and limit its further spread.

Future aspects

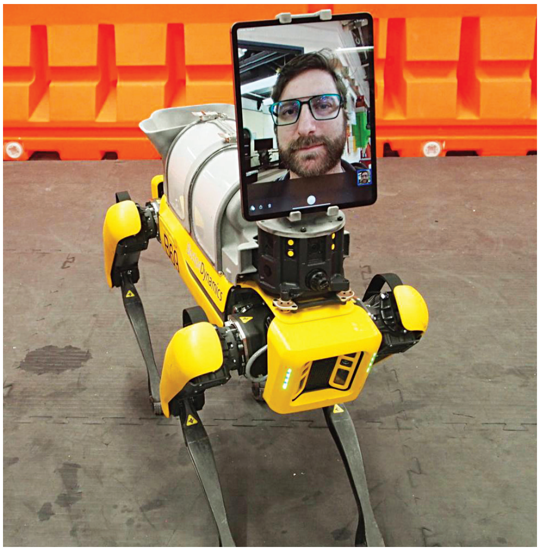
Healthcare professionals and medical scientists around the world are racing to gather and analyse the massive amounts of data in an effort to develop vaccines and treatments to bring the pandemic to an end. To fight the Covid-19 pandemic, healthcare delivery systems need the support of technologies such as supercomputers, cloud computing, artificial intelligence, machine learning, and geographic information system. Biotech as well as pharmaceutical organisations need extensive software support in terms of writing algorithm programs, developing software systems, and managing databases to develop suitable treatments and vaccines.

Healthcare organisations need these tools to make decisions faster, identify early infections, monitor the condition of the infected patients, provide improved treatment, and develop effective treatments and vaccines. Towards this direction, Covid-19 High-Performance Computing Consortium, a public-private joint effort, has been launched in the US.

In India, things are moving faster now under the National Supercomputing Mission launched in 2015. The mission is going to establish a network of supercomputers ranging from a few teraflops (TF) to hundreds of TFs and systems with greater than or equal to three petaflops (PF) in academic and research institutions by 2022. With three more supercomputers coming up, one each at IIT-Kanpur, JN Centre for Advanced Scientific Research, Bengaluru, and IIT-Hyderabad, the supercomputing facility will be ramped up to six PF. Eleven new systems are planned across India with a cumulative capacity of 10.4PF.

As per medical specialists, Covid-19 is here to stay, and future research in IT will play a vital role in learning, evolving, protecting, and moving towards positive results. END

Drones are being deployed for disinfection and for food and medicine delivery during Covid-19 outbreak (Credit: https://www.gsma.com)



Spot, a four-legged robot, being used by a doctor to treat Covid-19 patients remotely (*Credit: https://www.npr.org*)

Apart from research, information technology is being used to fight Covid-19 in several innovative ways (Credit: *www.rand.org)*

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The article was originally published in the March 2021 issue of Electronics For You.